

Multimodal Articulatory Inference: Radio-Visual Sensing for Silent Speech Recognition with RVTALL

Arham Siddiqui, Kaho Furukawa, Roy Yao, Akshay Rao, Tvisha Nepani

Software Division, Neurotech@Berkeley

Corresponding Lead: Tvisha Nepani, tnepani@berkeley.edu



ABSTRACT

Objective: Silent speech interfaces aim to restore communication for individuals who cannot speak by decoding intended speech from non-vocalized biosignals. We investigated whether silent speech decoding is possible by aligning and analyzing the following six sensor modalities collected during spoken utterances: radar, video, thermal imaging, laser distance sensing, EMG, and audio.

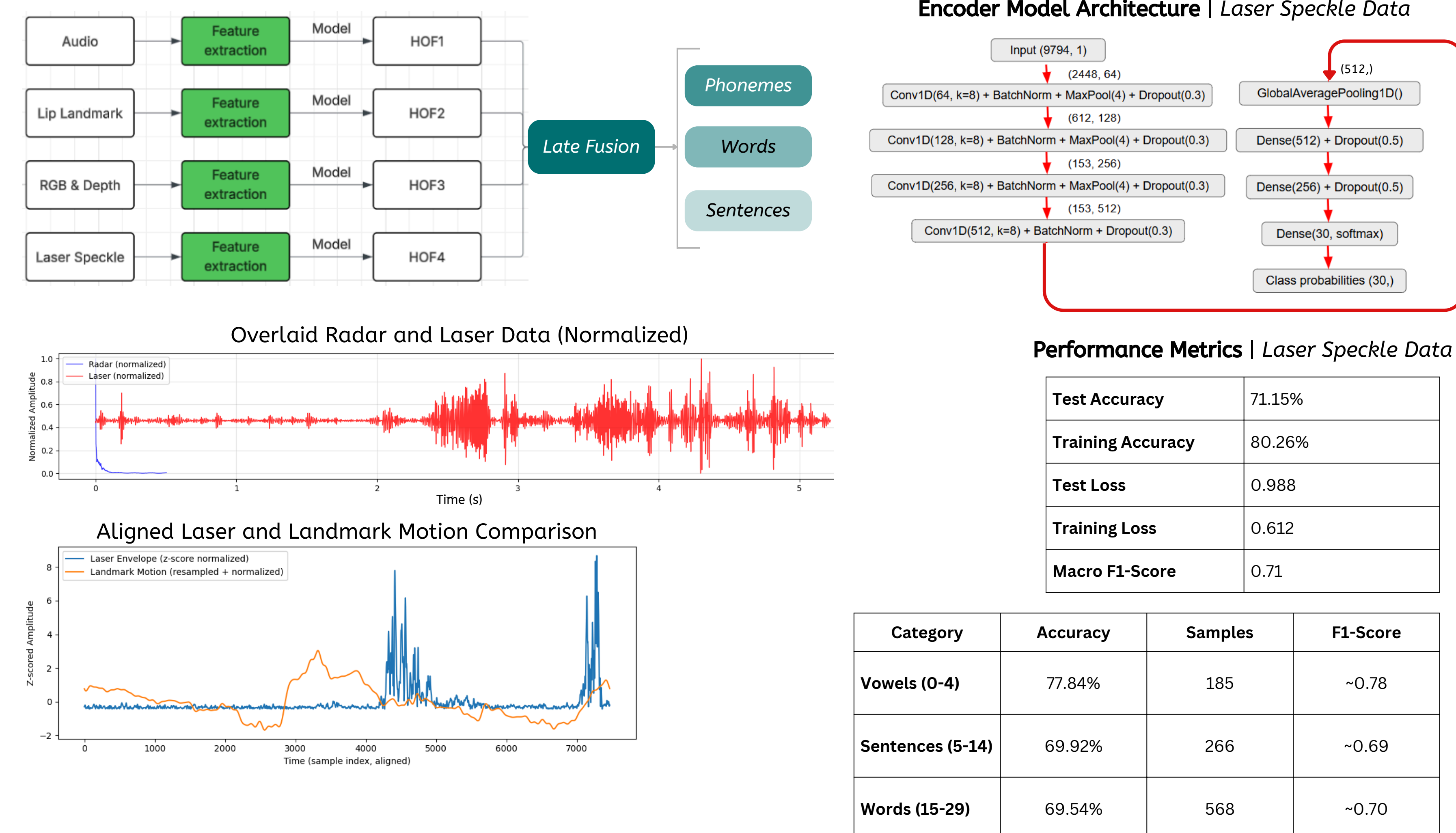
Methods: We initially aimed to time-align all raw sensor modalities, but our cross-modal visualizations showed that precise alignment was infeasible. We pivoted toward within-modality feature extraction, generating interpretable features for the six modalities. Using these features, we trained preliminary encoding models for each modality to understand which modality carries the most discriminative information.

Results: We achieved **71.15%** accuracy across 30 classes of vowels, words, and sentences using laser speckle data. This is ~20 times better than random guessing, which is around 3.33%. Our work establishes a foundation for silent-speech decoding through aligned data, feature extraction utilities, and analysis workflows.

METHODOLOGY

- Data Sourcing**
Decided to use the RVTALL dataset which contains 6 modalities (UWB radar, mmWave FMCW radar, laser speckle, RGB + depth/infrared, audio, lip/face landmarks)
- Data Synchronization**
Ensured each data modality was synchronized with the others by plotting samples and comparing peaks (see figures).
- Feature Extraction (radar data omitted)**
 - Audio: Mel-spectrogram, MFCCs, Pitch
 - Lip landmarks: Lip opening height/width, velocity & acceleration
 - RGB/Depth: CNN embeddings
 - Laser Speckle: Since no visual patterns are observed across the dataset, no feature extraction is used for speckle data.
- Developing Encoder Models:**
 - Generated modality-specific embeddings that will later be fused

DATA PROCESSING & ML PIPELINE



FEATURE EXTRACTION & RESULTS

A 1D CNN was used to learn the hierarchical temporal patterns for raw laser speckle time-series data. The training data expanded from 9,794 to 20,360 samples across 30 classes using data augmentation. The CNN uses 4 stacked convolutional blocks to extract features. In the end, global average pooling compresses the learned 512-d feature maps to a fixed 512-d vector and uses dense layers to map the probabilities over 30 classes.

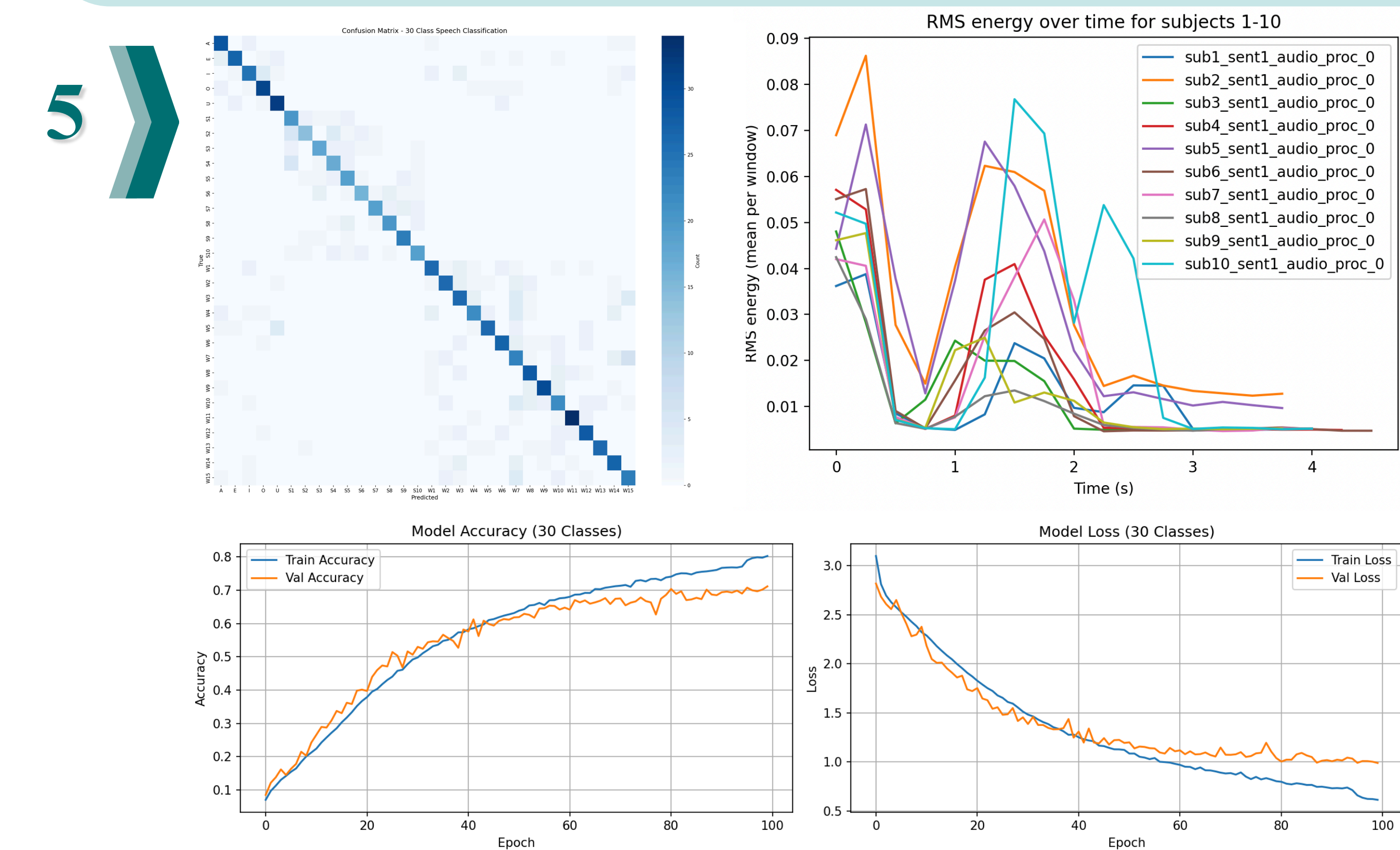


Figure 1: 30x30 confusion matrix showing predicted vs. true class labels. Darker diagonal elements indicate correct predictions; off-diagonal entries reveal systematic confusions. Figure 3: Training and validation accuracy (left) and loss curves (right) for the 30-class 1D CNN over 100 epochs.

INTRODUCTION & BACKGROUND

Silent speech interfaces (SSIs) aim to decode speech content without requiring audible acoustics, enabling communication for individuals who cannot vocalize.

- Early work from Denby et al. highlights the limitations of single-modality systems like lip video, EMG, ultrasound, and radar. They argue that future SSIs should shift toward **multimodal sensing** and feature-level representations to achieve usable results.
- Prior systems relying on video, throat EMG, or ultrasound alone struggle with variability in articulation patterns, limited generalization, and incomplete capture of internal articulatory dynamics.

Multimodal: where complementary modalities jointly capture both surface and subsurface articulator movement.

- Why RVTALL? Fused sensing overcomes the weaknesses of individual channels** = reliable, generalizable silent-speech decoding.

FUTURE DIRECTIONS

After training individual encoder models for each modality, we will be building a **late-fusion architecture** integrating modality-specific embeddings into a unified latent space for decoding.

We will also look into cross-modal complementarity, using attention-based gating / weighted concatenation so the **decoder can dynamically emphasize modalities that are cleanest for each articulation segment** (ex., radar during fast tongue movements, RGB during wide-mouth vowels).

At last, we want to extend to **continuous phoneme-level decoding**, enabling open-vocabulary silent speech.

ACKNOWLEDGEMENTS & REFERENCES

We thank the executive team at Neurotechnology at UC Berkeley for providing valuable support and encouragement throughout the research process and the Jacobs Institute for Design and Innovation for providing a facility to conduct our research and work sessions. We thank our software Leadership team - Beatrice, Tvisha, and Jeremy for their technical guidance, dedication, and mentorship in shaping this project as well as all our peers who have contributed their insights along the way. Lastly, we would like to acknowledge the University of California, Berkeley for funding this research, and for providing a platform to share and refine our research within the broader neurotech community. Their efforts in advancing the field continue to inspire our work.

- Tiwari, R., Sharma, R., Liang, W., Poppe, R., & Wiering, M. (2023). RVTALL: a dataset for learning radio-aided visual lip reading. Scientific Data, 10, 768. <https://doi.org/10.1038/s41597-023-02793-w>
- Denby, B., Schultz, T., Honda, K., Hueber, T., Gilbert, J. M., & Brumberg, J. S. (2010). Silent speech interfaces. Speech Communication, 52(4), 270-287
- Preprocessing code was adapted from the public RVTALL repository ([G-Bob](#))